

1. INTRODUCTION (VERY IMPORTANT)

? “Tell me about yourself”

✓ Strong Answer (tailored to this role)

I’m Lawrence, a final-year Computer Science student with over 3 years of experience building full-stack applications.

I’ve worked extensively with Java and Spring Boot for backend development, building systems like ERP platforms, POS systems, and payment-integrated applications.

I’m particularly interested in backend engineering — designing scalable APIs, handling data, and building reliable systems. Recently, I’ve also been working with modern stacks like NestJS and React, but Java remains my strongest backend language.

I enjoy solving real-world problems, especially in areas like payments and system integrations, which aligns well with what Aparte is building in proptech.

I’m looking for an opportunity to grow in a production environment, contribute to real systems at scale, and learn from experienced engineers.

? “Why do you want to join Aparte?”

✓ Answer

I’m interested in Aparte because you’re solving real problems in property access and transactions in Africa — which is a space with massive impact.

Technically, I’m excited about the opportunity to work on production-grade systems, especially in areas like bookings and payments, which require strong backend reliability.

I also like that the role emphasizes clean architecture, microservices, and DevOps — these are areas I’ve been learning and want to deepen through real-world experience.

◆ 2. CORE JAVA QUESTIONS

? What is OOP? Explain its pillars.

✓ Answer

OOP is a programming paradigm based on objects.

The four pillars are:

- **Encapsulation** – hiding data using private fields and exposing via methods

- **Inheritance** – reusing code via parent-child classes
 - **Polymorphism** – same method behaving differently (overloading/overriding)
 - **Abstraction** – hiding implementation details using interfaces/abstract classes
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? Difference between ArrayList and LinkedList

✓ Answer

Feature	ArrayList	LinkedList
Structure	Dynamic array	Doubly linked list
Access	Fast ($O(1)$)	Slow ($O(n)$)
Insert/Delete	Slower	Faster
Memory	Less	More

? What are Streams in Java?

✓ Answer

Streams are used for functional-style operations on collections. They allow operations like filtering, mapping, and reducing.

Example:

```
list.stream()
    .filter(x -> x > 10)
    .map(x -> x * 2)
    .collect(Collectors.toList());
```

? Checked vs Unchecked Exceptions

✓ Answer

- **Checked** → Must be handled (e.g., IOException)
 - **Unchecked** → Runtime exceptions (e.g., NullPointerException)
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◆ 3. SPRING BOOT (HIGH PRIORITY)

? What is Spring Boot?

✓ Answer

Spring Boot is a framework that simplifies building Java applications by providing auto-configuration, embedded servers, and production-ready features.

? Explain REST API principles

✓ Answer

- Stateless
- Use HTTP methods (GET, POST, PUT, DELETE)
- Use proper status codes
- Resource-based URLs

Example:

GET /api/users

POST /api/users

? What is Dependency Injection?

✓ Answer

It's a design pattern where dependencies are injected into a class instead of being created inside it.

? What is Spring Data JPA?

✓ Answer

It simplifies database access by reducing boilerplate code using repositories.

Example:

```
public interface UserRepository extends JpaRepository<User, Long> {}
```

? How do you handle errors in Spring Boot?

✓ Answer

Using `@ControllerAdvice` and `@ExceptionHandler` to centralize error handling.

◆ 4. DATABASES (IMPORTANT)

? What is normalization?

✓ Answer

It's the process of structuring a database to reduce redundancy.

? Difference between SQL and NoSQL

✓ Answer

SQL	NoSQL
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Structured	Flexible
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Tables	Documents/Key-value
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ACID	BASE
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? What is indexing?

✓ Answer

Improves query performance by allowing faster lookups.

◆ 5. SYSTEM DESIGN (THEY MAY TEST THIS LIGHTLY)

? How would you design a booking system?

✓ Answer (simple but solid)

I would:

- Use REST APIs for booking operations

- Store bookings in PostgreSQL
 - Use transactions to prevent double booking
 - Add caching (Redis) for frequent reads
 - Use queues for async tasks (notifications, payments)
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? How do you handle high traffic?

✓ Answer

- Load balancing
 - Caching (Redis)
 - Database optimization
 - Horizontal scaling
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◆ 6. PROJECT QUESTIONS (VERY IMPORTANT)

They WILL ask this.

? “Tell me about a project you built”

✓ Answer (structure it like this)

I built a [project name], which is a [what it does].

I used:

- Spring Boot for backend
- REST APIs for communication
- MySQL/PostgreSQL for database

One challenge I faced was [problem], and I solved it by [solution].

This project helped me improve in API design, debugging, and handling real-world scenarios.

? “What challenges have you faced?”

✓ Answer

Pick something real:

- Payment integration issues
- Debugging APIs
- Handling async flows

Then explain:

- Problem → Approach → Solution → Lesson
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◆ 7. AGILE / TEAMWORK

? What is Agile?

✓ Answer

Agile is an iterative approach to software development with continuous feedback.

? What happens in a standup?

✓ Answer

- What I did yesterday
 - What I'm doing today
 - Blockers
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◆ 8. QUESTIONS YOU SHOULD ASK THEM (VERY IMPORTANT)

Ask at least 1–2:

✓ Good ones:

- “What does success look like for an intern in the first 3 months?”
 - “What does your backend architecture currently look like?”
 - “Will I be working on production systems or staging first?”
 - “How do you handle code reviews and mentorship?”
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FINAL TIPS (CRITICAL)

- Speak clearly, not fast
- Think before answering (2–3 seconds pause is okay)
- If you don't know something:

“I haven't worked with that directly, but here's how I'd approach it...”

- Emphasize:
 - Spring Boot
 - REST APIs
 - Problem solving
 - Real projects